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PARTS LIST

Semiconductors:

IC1 - HT12E encoder
IC2 - HD12D decoder
T1 - BC547 npn transistor
LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1 - 1-mega-ohm
R2 - 33-kilo-ohm
R3, R4 - 1-kilo-ohm

Capacitors:

C1, C3 - 100 μ F, 16V electrolytic
C2 - 0.01 μ F ceramic disk

Miscellaneous:

S1 - Push-to-on switch
S2 - On/off SPST switch
CON1, CON2 - 2-pin connector
TX1 - 433MHz RF transmitter
RX1 - 433MHz RF receiver
PZ1 - Buzzer
ANT.1, ANT.2 - 25-30cm wire

Optional:

RL1 - 5V, SPDT relay
D1 - 1N4007 rectifier diode
CON2, CON3 - 2-pin connector
- 230V based call bell

RX1, which then decodes the signals, sending a response to decoder IC2. Consequently, pin 17 of IC2 becomes high. Transistor T1 conducts, energising relay RL1. This, in turn, supplies power to the 230V AC call bell through the common and normally open (NO) contacts of the relay, resulting in a louder sound.

Construction and testing

An actual-size, single-side PCB for the transmitting unit is shown in Fig. 6, with its component layout in Fig. 7. Assemble the unit within an appropriate enclosure and affix it to one side of the gate.

An actual-size, single-side PCB for the receiving unit is shown in Fig. 8 along with its component layout in Fig. 9. Assemble this unit also within a suitable enclosure

and position it on the opposite side of the gate or inside the house from where the bell can be heard from anywhere. It can be kept anywhere within the range of RF transmitter. It is important to note that no PCB is designed for the optional circuit diagram.

Both units operate at 5V. Employ an antenna for both the transmitter and receiver, using approximately 15-20cm of single-strand hook-up wire.

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/live-diy-wireless-call-bell> **EFY**

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